Qi Wang

Simulation & Control Engineer

Incoming Mechanical Engineering Master's student at the University of Sydney with a strong foundation in vehicle dynamics, control systems, and mechanical design. Proficient in translating complex physical problems into robust engineering solutions using advanced simulation tools (MATLAB/Simulink, SolidWorks, ADAMS). Experienced in project leadership, 3D modeling, and algorithm development (PID, Fuzzy Logic, Genetic Algorithms) to optimize system performance.



2746783219@qq.com | 0432703190 | Sydney, Australia

EDUCATION

Master of Professional Engineering (Mechanical)
The University of Sydney

08/2026 - Present *Sydney, Australia*

Bachelor of Engineering, Traffic Equipment and Control Engineering
Southwest Jiaotong University

09/2021 - 06/2025 *Chengdu, China*

Courses

- Mechanical Design, Engineering Testing and Signal Analysis, Automatic Control Principles, Vehicle Dynamics, Control Circuit Design and Simulation (EDA).

WORK EXPERIENCE

Research Assistant Intern
Wuhu Ruiji Intelligent Manufacturing Co., Ltd.

07/2023 - 08/2023 *Wuhu, China*

Achievements/Tasks

- Conducted in-depth market research and competitive product analysis to identify industry trends and address technical bottlenecks.
- Utilized CAD software for precise component drafting and actively contributed to the design optimization of suspension torsion bars, enhancing product performance and reducing manufacturing costs.

Product R&D Engineer Intern
Sichuan Yu Suan Ni Technology Co., Ltd.

07/2022 - 08/2022 *Sichuan, China*

Achievements/Tasks

- Developed detailed 3D models of mechanical components using SolidWorks based on 2D technical drawings and CAD files, ensuring strict adherence to engineering requirements.
- Enhanced model realism and visual appeal by applying advanced texturing techniques using Adobe Photoshop. Collaborated closely with the development team to adjust and optimize 3D models, ensuring seamless integration into final project deliverables.

SKILLS

SolidWorks (3D Modeling & Motion)

MATLAB / Simulink | CarSim | ADAMS

PID & Fuzzy Logic Control | Vehicle Dynamics Analysis

C++ & Data Analysis

PERSONAL PROJECTS

Advanced Vehicle Semi-Active Suspension Control Simulation (7-DOF) (12/2024 - 05/2025)

- Developed and validated advanced semi-active suspension control algorithms using a CarSim-Simulink co-simulation platform.
- Built high-fidelity vehicle dynamics models ranging from 1/4 car to 7-DOF full-car models.
- Designed and simulated multiple control strategies, including traditional PID, Skyhook, Fuzzy PID, and a Genetic Algorithm (GA) optimized PID controller.
- Achieved a 58.7% reduction in sprung mass vertical acceleration RMS compared to passive suspension using the GA-PID strategy, significantly enhancing ride comfort while maintaining handling stability.

CERTIFICATES

500px contracted photographer

LANGUAGES

Chinese
Native or Bilingual Proficiency

English
Professional Working Proficiency

German
Professional Working Proficiency

INTERESTS

Photography | Travel